

Technology Stack

Date	2 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML	Used to build the Home Page, User Input Forms, and Prediction Result Display pages of the Flask web application

2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python, Flask	Flask is used to develop the interactive web application. Python handles all data processing, model training, and prediction logic.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	Google Sheets	The loan eligibility dataset containing features like Gender, Education, Applicant Income, and Loan Amount is downloaded from Google Sheets.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Hugging Face Spaces	The completed web application is deployed and hosted on Hugging Face Spaces for public access.
5	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, SciPy	NumPy and Pandas handle data processing. Matplotlib and Seaborn generate visualizations. Scikit-learn is used to build and train the classification models (Decision Tree, Random Forest, KNN, XGBoost).